

D3 — top 5 fixes before this runs unattended, ranked by impact if skipped

- 1. **The batch conflict has no resolution path.** 2024-01 and 2024-03 each arrived twice, 234 rows, no timestamp or revision flag to arbitrate. Pipeline defaults to DEFAULT-BATCH-01 and marks every derived figure provisional — honest, not useful, at 100% provisional. *Fix:* a triage queue where a named analyst records the choice and figures promote to settled (specified, not built).
- 2. **Ingestion assumes one file shape.** The 71-field crosswalk was learned by hand; real DHIS2, HealthTrack, OpenMRS and paper forms won't hold still. *Fix:* LLM-proposed, human-confirmed, cached column mapping — never re-inferred per run. Free win: HXL-tagged sources map by lookup, no model call needed.
- 3. **Chart runtime decided for the bulletin, open for the surface that doesn't exist yet.** Flint → Vega-Lite adopted and working. *Do not* default the unbuilt explore surface to whichever runtime is closest to hand — decide explicitly, in writing, before its first line is written.
- 4. **Nothing runs on a schedule, no one is told when it fails.** Hamilton already exposes the DAG as a callable — orchestration is a wrapper, not a rewrite. What matters more than the scheduler: publication must gate on the 5 checks passing, and a failed run must reach a person.
- 5. **Facility geography is unusable.** GPS is uniform-random inside Rwanda's bounding box for all 117 facilities; the bulletin honestly maps at district grain. *Fix:* join HDX's Rwanda Healthsites layer (1,345 real facilities, open licence). *Do not* reverse-geocode district centroids as a substitute — that manufactures false precision.

D4 — Handover

The deliverable is not the artifact, it's a named Digital Health Officer independently restarting the pipeline once, unassisted, before exit. Two training sessions: run the full cycle hands-on-keyboard-DHO; diagnose a deliberately seeded failure using only the runbook. Full clean-rebuild verified 2026-08-11 from a wiped state — 5/5 checks passed, output reproduced at the same byte hash, confirming the pipeline is deterministic. That run predates later pipeline changes (ADR-0012) and has **not been re-verified against current code** — flagged here, not hidden; re-running and diff-checking the hash is the next action before reproducibility can be re-claimed.

Three documents answer "what would you hand the MoH IT team" and ship in the repository, not this PDF — at nine pages combined they would have made D3+D4 a third of a document whose brief asks for 80% of the effort on D1+D2:

deliverable-4-handover/runbook.md · data-contract.md · decision-register.md

Full detail for all four deliverables, the pipeline code, and the openspec behaviour specs: github.com/nyashkn/sand-fde-int